

4.2 Mean and variance of quadratic forms

4.2.1 The mean needs no normality

Theorem 2.3.1 computed $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ for any random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$:

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}. \quad (4.2.1)$$

The formula has two parts with distinct roles. The trace term is what the noise contributes. The term $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ is what the signal contributes. For nonnegative definite $\mathbf{A}$ it vanishes exactly when $\mathbf{A} \boldsymbol{\mu} = \mathbf{0}$, which describes the mean vectors a well-designed estimator is meant to ignore.

Example. Two estimators of σ^2 under a drifting mean

Let $Y_1, \dots, Y_n$ be uncorrelated with common variance σ^2 , and consider

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad Q = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2,$$

the sample variance and the **mean square successive difference**. In matrix form $S^2 = \mathbf{Y}^\top \mathbf{C} \mathbf{Y} / (n-1)$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$, and $Q = \mathbf{Y}^\top \mathbf{D}^\top \mathbf{D} \mathbf{Y} / \{2(n-1)\}$, where $\mathbf{D}$ is the $(n-1) \times n$ first-difference matrix with rows $(\dots, -1, 1, \dots)$. Both matrices annihilate $\mathbf{1}$, so both estimators ignore a constant mean. Since $\text{tr}(\mathbf{C}) = n-1$ and $\text{tr}(\mathbf{D}^\top \mathbf{D}) = \text{tr}(\mathbf{D} \mathbf{D}^\top) = 2(n-1)$, Eq. 4.2.1 with $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ shows that both are unbiased when the mean is constant.

Now let the mean drift linearly, $\mu_i = \mu_0 + \theta i$. Then $\mathbf{C} \boldsymbol{\mu}$ is the centred trend and $\mathbf{D} \boldsymbol{\mu} = \theta \mathbf{1}_{n-1}$, so

$$E(S^2) = \sigma^2 + \theta^2 \frac{n(n+1)}{12}, \quad E(Q) = \sigma^2 + \frac{\theta^2}{2}.$$

The bias of S^2 grows quadratically with the length of the series, while that of Q does not grow at all. With $n = 20$, $\sigma = 1$ and $\theta = 0.1$, the means are 1.350 and 1.005. Differencing removes smooth trends, and this is why Q was proposed for series whose level may wander (von Neumann 1941). The price of robustness to drift is paid in variance, which we compute below.

4.2.2 A canonical representation

Higher moments and distributions need normality, and one representation answers all such questions at once.

Theorem 4.2.1. Canonical form of a normal quadratic form

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and let $\mathbf{A}$ be symmetric. Write $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$ with $\mathbf{L}$ of size $n \times k$, and take a spectral decomposition $\mathbf{L}^\top \mathbf{A} \mathbf{L} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$ with $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$. Put $\mathbf{b} = \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \sum_{j=1}^k (\lambda_j W_j^2 + 2b_j W_j), \quad \mathbf{W} \sim N_k(\mathbf{0}, \mathbf{I}_k). \quad (4.2.2)$$

For $\lambda_j \neq 0$ the j th term equals $\lambda_j(W_j + b_j/\lambda_j)^2 - b_j^2/\lambda_j$. The nonzero λ_j are the nonzero eigenvalues of $\mathbf{A}\Sigma$.

Proof. Such an $\mathbf{L}$ exists. By Theorem 1.7.2, $\Sigma = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ orthogonal and $\mathbf{D}$ diagonal with entries $d_i \geq 0$, and $\mathbf{L} = \mathbf{U}\mathbf{D}^{1/2}$ works with $k = n$. Let $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$. The vector $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ is normal with mean $\boldsymbol{\mu}$ and covariance $\mathbf{L}\mathbf{L}^\top = \Sigma$ (Theorem 3.3.1). A normal distribution is determined by its mean and covariance (Theorem 3.2.1), so $\mathbf{Y}$ and $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ have the same law, and so do $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ and

$$(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z})^\top \mathbf{A}(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + 2\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} + \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z}.$$

Put $\mathbf{W} = \mathbf{Q}^\top \mathbf{Z}$, again $N_k(\mathbf{0}, \mathbf{I})$. Then $\mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{W}^\top \boldsymbol{\Lambda} \mathbf{W} = \sum \lambda_j W_j^2$ and $\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{b}^\top \mathbf{W}$, which is Eq. 4.2.2. The completed square is algebra. Finally, $\mathbf{L}^\top \mathbf{A}\mathbf{L} = \mathbf{H}\mathbf{F}$ with $\mathbf{F} = \mathbf{A}\mathbf{L}$ and $\mathbf{H} = \mathbf{L}^\top$, and $\mathbf{F}\mathbf{H} = \mathbf{A}\Sigma$. By Proposition 1.7.1(d) these two products have the same nonzero eigenvalues, with the same multiplicities. ■

So a normal quadratic form is a constant plus a sum of independent pieces. Each piece is either a scaled noncentral $\chi^2(1)$ variable, with scale λ_j , or a normal variable with mean zero, when $\lambda_j = 0$ but $b_j \neq 0$. Because the λ_j may be negative, $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$ need not be positive. The linear pieces are special to singular covariances. If Σ is positive definite, $\mathbf{L}$ can be taken square and nonsingular, and then $\lambda_j = 0$ gives $\mathbf{A}\mathbf{L}\mathbf{q}_j = \mathbf{0}$ for the j th column $\mathbf{q}_j$ of $\mathbf{Q}$, so $b_j = (\mathbf{A}\mathbf{L}\mathbf{q}_j)^\top \boldsymbol{\mu} = 0$. A linear piece needs a direction in which $\mathbf{A}$ responds to the mean while the noise is absent.

Corollary 4.2.2. Moment generating function

In the setting of Theorem 4.2.1, for $|t| < 1/(2 \max_j |\lambda_j|)$ (any t if all $\lambda_j = 0$),

$$\mathbb{E} e^{t\mathbf{Y}^\top \mathbf{A}\mathbf{Y}} = e^{t\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}} \prod_{j=1}^k (1 - 2\lambda_j t)^{-1/2} \exp\left(\frac{2b_j^2 t^2}{1 - 2\lambda_j t}\right).$$

Proof. The terms of Eq. 4.2.2 are independent. If $\lambda_j = 0$, the term $2b_j W_j$ has mgf $e^{2b_j^2 t^2}$. If $\lambda_j \neq 0$, apply Lemma 4.1.1 to $W_j + \nu_j$ with $\nu_j = b_j/\lambda_j$ at the argument $\lambda_j t$, and multiply by $e^{-tb_j^2/\lambda_j}$. The exponent is $b_j^2 t / \{\lambda_j(1 - 2\lambda_j t)\} - b_j^2 t / \lambda_j = 2b_j^2 t^2 / (1 - 2\lambda_j t)$. ■

4.2.3 Variance and covariances

Theorem 4.2.3. Moments of normal quadratic forms

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$, let $\mathbf{A}$ and $\mathbf{B}$ be symmetric $n \times n$ matrices, and let $\mathbf{K}$ be an $m \times n$ matrix. Then

$$(a) \quad \mathbb{E}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu};$$

- (b) $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \text{tr}\{(\mathbf{A} \boldsymbol{\Sigma})^2\} + 4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu};$
 (c) $\text{Cov}(\mathbf{K} \mathbf{Y}, \mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \mathbf{K} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu};$
 (d) $\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\Sigma}) + 4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{B} \boldsymbol{\mu}.$

Proof.

- (a) is Eq. 4.2.1. For (b), use Eq. 4.2.2. Independent terms have additive variances, and for $W \sim N(0, 1)$, $\text{Var}(\lambda W^2 + 2bW) = 2\lambda^2 + 4b^2$, because $\text{Cov}(W^2, W) = E W^3 = 0$. Hence

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \sum_j \lambda_j^2 + 4 \|\mathbf{b}\|^2 = 2 \text{tr}\{(\mathbf{L}^\top \mathbf{A} \mathbf{L})^2\} + 4 \boldsymbol{\mu}^\top \mathbf{A} \mathbf{L} \mathbf{Q} \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}.$$

Now $\text{tr}\{(\mathbf{L}^\top \mathbf{A} \mathbf{L})^2\} = \text{tr}(\mathbf{L}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \mathbf{L}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$ by Theorem 1.6.1, and $\mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$, so the second term is $4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$. For (c), write $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{L} \mathbf{Z}$ as in the proof of Theorem 4.2.1. Then $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} - E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A} \mathbf{L} \mathbf{Z} - \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + 2 \boldsymbol{\mu}^\top \mathbf{A} \mathbf{L} \mathbf{Z}$ and $\mathbf{K} \mathbf{Y} - E(\mathbf{K} \mathbf{Y}) = \mathbf{K} \mathbf{L} \mathbf{Z}$. Every entry of $E[\mathbf{Z}(\mathbf{Z}^\top \mathbf{G} \mathbf{Z})]$ is a sum of third moments of a $N_k(\mathbf{0}, \mathbf{I})$ vector, which vanish. So only the linear term contributes, and $E[\mathbf{K} \mathbf{L} \mathbf{Z} \cdot 2 \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}] = 2 \mathbf{K} \mathbf{L} \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$. For (d), apply (b) to $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{A} + \mathbf{B}$, and use $\text{Var}(X + Y) = \text{Var} X + \text{Var} Y + 2 \text{Cov}(X, Y)$. ■

For $\boldsymbol{\mu} = \mathbf{0}$, part (b) can also be read off the fourth moments of a normal vector (Exercise (B1)). The variance has the same two-part structure as the mean: a pure-noise term, plus a term that is present only when $\mathbf{A}$ responds to the mean. Part (c) shows that a quadratic form and a linear form are uncorrelated when $\mathbf{K} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{0}$. Under normality this will turn out to give independence (Theorem 4.4.1).

Example

Example (Two estimators of σ^2 under a drifting mean)

With $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ and a constant mean, Theorem 4.2.3(b) gives $\text{Var}(S^2) = 2\sigma^4 \text{tr}(\mathbf{C}^2)/(n-1)^2 = 2\sigma^4/(n-1)$. For Q , the matrix $\mathbf{D} \mathbf{D}^\top$ is tridiagonal with 2 on the diagonal and -1 beside it, so $\text{tr}\{(\mathbf{D}^\top \mathbf{D})^2\} = \text{tr}\{(\mathbf{D} \mathbf{D}^\top)^2\} = 4(n-1) + 2(n-2) = 6n-8$ and

$$\text{Var}(Q) = \frac{2\sigma^4(6n-8)}{4(n-1)^2} = \frac{\sigma^4(3n-4)}{(n-1)^2}.$$

The ratio $\text{Var}(Q)/\text{Var}(S^2) = (3n-4)/\{2(n-1)\}$ tends to $3/2$. With $n = 20$ the two variances are 0.1551 and 0.1053, a ratio of 1.474. So with no drift, Q needs about half as many observations again to match S^2 . Under the drift $\theta = 0.1$, the signal term $4 \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$ raises the variances to 0.1552 and 0.1789. Combining variance and bias, the root mean squared errors are 0.394 for Q and 0.549 for S^2 . The listing evaluates the formulas, and the script confirms all eight moments with 400,000 simulated series.

```
import numpy as np
rng = np.random.default_rng(4402)
n, sigma, theta = 20, 1.0, 0.1
t = np.arange(1, n + 1)
mu = 5.0 + theta * t
```

mean with a linear trend

```

Sigma = sigma**2 * np.eye(n)

def qf_moments(A, mu, Sigma):
    """Mean and variance of  $Y^T A Y$  for  $Y \sim N(\mu, \text{Sigma})$ ,  $A$  symmetric."""
    AS = A @ Sigma
    mean = np.trace(AS) + mu @ A @ mu
    var = 2 * np.trace(AS @ AS) + 4 * mu @ AS @ A @ mu
    return mean, var

C = np.eye(n) - np.ones((n, n)) / n          # centring matrix
D = np.diff(np.eye(n), axis=0)               # (n-1) x n first differences
A_s2 = C / (n - 1)
A_q = D.T @ D / (2 * (n - 1))
for name, A in [("S2", A_s2), ("Q", A_q)]:
    print(name, qf_moments(A, mu, Sigma), qf_moments(A, 0 * mu, Sigma))

```

[Run this code online](#)

Normality enters (b)–(d) through the third and fourth moments of $\mathbf{Y}$. For independent coordinates with excess kurtosis the variance acquires an extra term that involves only the diagonal of $\mathbf{A}$ (Theorem 2.3.3). Variance formulas for sums of squares derived under normality can be badly wrong for heavy-tailed errors, a point taken up in [Chapter 19](#) (Proposition 19.3.4, Proposition 19.3.5).

4.2.4 Exercises

4.2.4.1 A. Check your understanding

Exercise. A1

Let $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ and $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Compute $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ directly, and compare with $2 \text{tr}(\mathbf{A}^2)$ and with $2 \text{tr}(\mathbf{A}_s^2)$, where $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$.

4.2.4.2 B. Practice

Exercise. B1

Let (U_i, W_i) , $i = 1, \dots, n$, be independent copies of a bivariate normal pair with variances σ_U^2 , σ_W^2 and covariance σ_{UW} , and let $s_{UW} = (n-1)^{-1} \sum_i (U_i - \bar{U})(W_i - \bar{W})$. Write $s_{UW} = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ for the stacked vector $\mathbf{Y} = (\mathbf{U}^\top, \mathbf{W}^\top)^\top$ and a symmetric $2n \times 2n$ matrix $\mathbf{A}$, and use [Theorem 4.2.3](#) to show that $\text{Var}(s_{UW}) = (\sigma_U^2 \sigma_W^2 + \sigma_{UW}^2) / (n-1)$. Check the answer against $\text{Var}(S^2)$ when $\mathbf{U} = \mathbf{W}$.

Exercise. B2

Let $Y_1, \dots, Y_n$ be independent with mean 0, variance σ^2 and $E Y_i^4 = \mu_4$. By Exercise (B4), for symmetric $\mathbf{A}$ and $\mathbf{B}$,

$$\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B}).$$

Show that in a one-way layout with $g \geq 2$ groups, not all of size one, the between-group and within-group sums of squares are correlated unless $\mu_4 = 3\sigma^4$. Which sign does the correlation

have for heavy-tailed errors?

Solution. In the one-way layout the between-group matrix is $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$ and the within-group matrix is $\mathbf{B} = \mathbf{I} - \mathbf{M}$. They satisfy $\mathbf{AB} = \mathbf{0}$, so $\text{tr}(\mathbf{AB}) = 0$ and the covariance reduces to $(\mu_4 - 3\sigma^4) \sum_i a_{ii}b_{ii}$. An observation in group ℓ has $a_{ii} = 1/n_\ell - 1/n$ and $b_{ii} = 1 - 1/n_\ell$, so

$$\sum_i a_{ii}b_{ii} = \sum_\ell n_\ell \left(\frac{1}{n_\ell} - \frac{1}{n} \right) \left(1 - \frac{1}{n_\ell} \right) = \sum_\ell \left(1 - \frac{n_\ell}{n} \right) \left(1 - \frac{1}{n_\ell} \right).$$

With $g \geq 2$ every first factor is positive, and the second is positive for any group with two or more observations. So the sum is positive, and the covariance vanishes iff $\mu_4 = 3\sigma^4$. For heavy-tailed errors, $\mu_4 > 3\sigma^4$, the two sums of squares are positively correlated: a large error inflates both.

Exercise. B3

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{M}$ be symmetric idempotent of rank r with $\mathbf{M}\boldsymbol{\mu} \neq \boldsymbol{\mu}$, as when a regression model omits a needed regressor. Show that $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/\sigma^2)$, and find the bias of $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - r)$ as an estimator of σ^2 . In the setting of Theorem 4.4.6 without the assumption $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$, which parts of the conclusion survive?

Exercise. B4

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$. Use Corollary 4.2.2 and Proposition 1.7.1(d) to show that $E e^{t\mathbf{Y}^\top \mathbf{A} \mathbf{Y}} = \det(\mathbf{I}_n - 2t\mathbf{A}\boldsymbol{\Sigma})^{-1/2}$ for $|t|$ small.

4.2.4.3 C. Going deeper

Exercise. C1

Let $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$ with μ and σ^2 unknown. Among symmetric $\mathbf{A}$ with $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$ for all μ and σ^2 , show that $\mathbf{A} = \mathbf{C}/(n - 1)$, which gives S^2 , uniquely minimizes $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ at every parameter value. (Compare Exercise (B5), where the mean is known to be zero.)

Solution. Since $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr} \mathbf{A} + \mu^2 \mathbf{1}^\top \mathbf{A} \mathbf{1}$, unbiasedness means $\text{tr} \mathbf{A} = 1$ and $\mathbf{1}^\top \mathbf{A} \mathbf{1} = 0$. By Theorem 4.2.3, $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\mu^2 \sigma^2 \|\mathbf{A} \mathbf{1}\|^2$. The map $\mathbf{A} \mapsto \mathbf{C} \mathbf{A} \mathbf{C}$ is an orthogonal projection for the inner product $\text{tr}(\mathbf{A} \mathbf{B})$ on symmetric matrices, so $\text{tr}(\mathbf{A}^2) \geq \text{tr}\{(\mathbf{C} \mathbf{A} \mathbf{C})^2\}$, with equality iff $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$. Also $\text{tr}(\mathbf{C} \mathbf{A} \mathbf{C}) = \text{tr} \mathbf{A} - \mathbf{1}^\top \mathbf{A} \mathbf{1}/n = 1$. The symmetric matrix $\mathbf{C} \mathbf{A} \mathbf{C}$ has rank at most $n - 1$. If $\lambda_1, \dots, \lambda_{n-1}$ are its eigenvalues on $\mathbf{1}^\perp$, the Cauchy-Schwarz inequality gives $1 = (\sum \lambda_i)^2 \leq (n - 1) \sum \lambda_i^2$. Hence $\text{tr}(\mathbf{A}^2) \geq 1/(n - 1)$, with equality iff $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$ and all $\lambda_i = 1/(n - 1)$, that is, iff $\mathbf{A} = \mathbf{C}/(n - 1)$. That matrix also has $\mathbf{A} \mathbf{1} = \mathbf{0}$, so it minimizes both variance terms at once.